# Robotic Manipulation

***Perception, Planning, and Control***

Russ Tedrake


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How to cite these notes, use annotations, and give feedback.

**Note:** These are working notes used for a course being taught at MIT. They will be updated throughout the Fall 2024 semester.

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# Appendix A
# Spatial Algebra

Throughout these notes we've introduced the basic rules of spatial algebra. I find myself looking back at them often! For ease of reference, I've compiled them again here. These make heavy use of our "monogram notation".

The Drake documentation also has a very nice summary of the multibody quantities and how they map to the monogram notation.

## A.1 Position, Rotation, and Pose

As we introduced here, we use ${}^Bp^A_C$ to denote a position of point or frame $A$ relative to point or frame $B$ expressed in frame $C$. We use ${}^BR^A$ to denote the orientation of frame $A$ measured from frame $B$; unlike vectors, pure rotations do not have an additional "expressed in" frame. Similarly, we use ${}^BX^A$ to denote the pose/transform of frame $A$ measured from frame $B$. We do not use the "expressed in" frame subscript for pose; we always want the pose expressed in the reference frame.

The basic rules of spatial algebra are as follows:

- Positions expressed in the same frame can be added when their reference and target symbols match:

  $$ {}^Ap^B_F + {}^Bp^C_F = {}^Ap^C_F. \tag{1}$$

  Addition is commutative, and the additive inverse is well defined:

  $$ {}^Ap^B_F = -{}^Bp^A_F. \tag{2}$$

  Those should be pretty intuitive; make sure you confirm them for yourself.
- Multiplication by a rotation can be used to change the "expressed in" frame:

  $$ {}^Ap^B_G = {}^GR^F\,{}^Ap^B_F. \tag{3}$$

  You might be surprised that a rotation alone is enough to change the expressed-in frame, but it's true. The position of the expressed-in frame does *not* affect the relative position between two points.
- Rotations can be multiplied when their reference and target symbols match:

$$ {}^AR^B \, {}^BR^C = {}^AR^C. \tag{4} $$

The inverse operation is also simply defined:

$$ \left[{}^AR^B\right]^{-1} = {}^BR^A. \tag{5} $$

When the rotation is represented as a rotation matrix, this is literally the matrix inverse, and since rotation matrices are orthonormal, we also have $R^{-1} = R^T$.

- Transforms bundle this up into a single, convenient notation when positions are measured from a frame (and the same frame they are expressed in):

$$ {}^Gp^A = {}^GX^F \, {}^Fp^A = {}^Gp^F + {}^Fp^A_G = {}^Gp^F + {}^GR^F \, {}^Fp^A. \tag{6} $$

- Transforms compose:

$$ {}^AX^B \, {}^BX^C = {}^AX^C, \tag{7} $$

and have an inverse

$$ \left[{}^AX^B\right]^{-1} = {}^BX^A. \tag{8} $$

Please note that for transforms, we generally do *not* have that $X^{-1}$ is $X^T$, though it still has a simple form.

## A.2 Spatial velocity

As we introduced here, we represent the rate of change in pose using a six-component vector for *spatial velocity*:

$$ {}^AV^B_C = \begin{bmatrix} {}^A\omega^B_C \\ {}^A\mathrm{v}^B_C \end{bmatrix}. \tag{9} $$

${}^AV^B_C$ is the spatial velocity (also known as a "twist") of frame $B$ measured in frame $A$ expressed in frame $C$, ${}^A\omega^B_C \in \mathbb{R}^3$ is the *angular velocity* (of frame $B$ measured in $A$ expressed in frame $C$), and ${}^A\mathrm{v}^B_C \in \mathbb{R}^3$ is the *translational velocity* (along with the same shorthands as for positions). Spatial velocities fit nicely into our spatial algebra:

- Velocities add when they are expressed in the same frame:

$$ {}^A\mathrm{v}^B_F + {}^B\mathrm{v}^C_F = {}^A\mathrm{v}^C_F, \qquad {}^A\omega^B_F + {}^B\omega^C_F = {}^A\omega^C_F, \tag{10} $$

and have the additive inverse, ${}^AV^C_F = -{}^CV^A_F$,.

- Rotations can be used to change between the "expressed-in" frames:

$$ {}^A\mathrm{v}^B_G = {}^GR^F \, {}^A\mathrm{v}^B_F, \qquad {}^A\omega^B_G = {}^GR^F \, {}^A\omega^B_F. \tag{11} $$

- Translational velocities compose across frames with:

$$ {}^A\mathrm{v}^C_F = {}^A\mathrm{v}^B_F + {}^B\mathrm{v}^C_F + {}^A\omega^B_F \times {}^Bp^C_F. \tag{12} $$

- This reveals that additive inverse for translational velocities is not obtained by switching the reference and measured-in frames; it is slightly more complicated:

$$ -{}^A\mathrm{v}^B_F = {}^B\mathrm{v}^A_F + {}^A\omega^B_F \times {}^Bp^A_F. \tag{13} $$

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## A.3 Spatial force

As we introduced here, we define a six-component vector for *spatial force*, using the monogram notation:

$$F^{B_p}_{\text{name},C} = \begin{bmatrix} \tau^{B_p}_{\text{name},C} \\ f^{B_p}_{\text{name},C} \end{bmatrix} \quad \text{or,} \quad \text{if you prefer} \quad \left[F^{B_p}_{\text{name}}\right]_C = \begin{bmatrix} \left[\tau^{B_p}_{\text{name}}\right]_C \\ \left[f^{B_p}_{\text{name}}\right]_C \end{bmatrix}. \tag{14}$$

$F^{B_p}_{\text{name},C}$ is the named spatial force applied to a point, or frame origin, $B_p$, expressed in frame $C$. The name is optional, and the expressed in frame, if unspecified, is the world frame. For forces in particular, it is recommended that we include the body, $B$, that the force is being applied to in the symbol for the point $B_p$, especially since we will often have equal and opposite forces.

Spatial forces fit neatly into our spatial algebra:

- Spatial forces add when they are applied to the same body in the same frame, e.g.:

$$F^{B_p}_{\text{total},C} = \sum_i F^{B_p}_{i,C}. \tag{15}$$

- Shifting a spatial force from one application point, $B_p$, to another point, $B_q$, uses the cross product:

$$f^{B_q}_C = f^{B_p}_C, \qquad \tau^{B_q}_C = \tau^{B_p}_C + {}^{B_q}p^{B_p}_C \times f^{B_p}_C. \tag{16}$$

- As with all spatial vectors, rotations can be used to change between the "expressed-in" frames:

$$f^{B_p}_D = {}^DR^C f^{B_p}_C, \qquad \tau^{B_p}_D = {}^DR^C \tau^{B_p}_C. \tag{17}$$